

Seismic Event Classification Based on Attention Mechanism and Spatial Pyramid Pooling

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Abstract—The precise classification of seismic events is of paramount importance for earthquake early warning, risk management, and seismic cataloging. A convolutional neural network (CNN) model is proposed, integrating short-time Fourier transform, attention mechanism and spatial pyramid pooling. The aim is to efficiently classify seismic events by capturing core features and multi-scale information. The proposed model is subject to experimental evaluation through five-fold cross-validation using a dataset collected and labeled by the Jiangsu Seismic Network Center, encompassing natural earthquakes, artificial explosions, and collapses. Compared to other methods, the proposed model exhibits a significant improvement in accuracy, ranging from 90.20% to 92.90%, reaching 94.18%. The recognition accuracy for natural earthquakes, collapses, and explosions is reported as 93.52%, 94.34%, and 94.80%, respectively. Optimal results are achieved across various performance metrics, including sensitivity (94.10%), specificity (97.05%), precision (94.22%), F1 score (93.15%) and Matthews correlation coefficient (91.24%). Comprehensive experimental results indicate that this method has made a significant breakthrough in the task of seismic event classification, providing a more robust tool for seismic research.

Keywords—Seismic event classification; Short-time Fourier transform; Deep learning; Spatial pyramid pooling; Attention mechanism

I. INTRODUCTION

The classification of seismic events holds significant importance for the scientific rigor and accuracy of seismic studies. However, the current classification methods heavily rely on manual intervention, which is time-consuming, inefficient, and prone to issues of misreporting. Therefore, there is an urgent need for more efficient and accurate automated classification methods to address this challenge.

A. Related work

To address this issue, comprehensive experiments and studies have been undertaken by researchers. Depending on the employed classification method, earthquake event classification can be primarily categorized into three methods: feature criteria, template matching, and deep learning.

The criteria for feature selection rely on empirical analysis, extracting relevant features from seismic event waveforms in the time and frequency domains for earthquake classification [1,2]. However, relying solely on a simple combination of individual or a few features lacks generality and is challenging to apply across multiple regions and categories. Template matching involves matching specific features of seismic events with pre-defined type templates to determine their similarity and subsequently classify them. Literature [3,4] has demonstrated its effectiveness, but this method still requires manual feature selection, making it difficult to find a widely applicable and effective combination of features and classifiers.

The advancements in artificial intelligence (AI) technology have led to the increasing utilization of deep learning methods in earthquake event classification, owing to the abundance of seismic data. Researchers have explored various approaches to improve classification accuracy and robustness. For instance, Mousavi et al. [5] integrated attention mechanisms and residual structures into long short-term memory networks, achieving high sensitivity in monitoring and phase picking of small earthquakes. Similarly, Miao et al. [6] utilized frequency domain amplitude spectrum information from multiple time windows and employed a fully connected neural network for classification, achieving an impressive accuracy of 97% in identifying earthquakes and blasting events. Additionally, Wei Yonggang et al. [7] employed deep learning residual networks with seismic signal power spectra as input, demonstrating high recognition rates and stable performance. Li Bingjun et al. [8] applied the short-time Fourier transform to generate time-frequency images for convolutional neural network input, yielding an average accuracy of 97.39%. Notably, the deep learning approach adopted in these studies eliminates the need for manual feature selection, thereby reducing subjectivity and enhancing the model's robustness.

Considering the substantial progress in AI research over the past 65 years, which has been recognized as a pivotal driver of socio-economic transformation, there is a growing interest in applying AI technologies to disciplines such as seismology [9]. Consequently, this paper aims to leverage AI methodologies in the field of seismology, particularly in earthquake

classification research, to provide a practical and effective tool for seismic event analysis.

B. Main work and innovation points

Previous research in the field of earthquake event classification has been criticized for its fixed receptive fields, information loss, and fixed weights. In response to these limitations, this study proposes a three-classification approach that incorporates multiscale information fusion and key feature extraction.

- Time-frequency feature extraction. Capturing the core features of seismic signals is achieved through short-time Fourier transform, enhancing the overall classification performance.
- Multi-scale information fusion. The spatial pyramid pooling structure is employed to replace global average pooling, comprehensively considering spatial information at different scales on seismic waveforms.
- Multi-dimensional feature integration. The integration of spatial attention mechanism and channel attention mechanism into the convolutional neural network enhances the model's comprehensiveness in information extraction.

II. DATASETS

The dataset employed in this study was collected, organized, and annotated by the Jiangsu Seismological Bureau, covering three primary categories of seismic events: explosions, collapses, and natural earthquakes.

The experimental results underwent rigorous evaluation via five-fold cross-validation, a method where the dataset was partitioned into five equally sized samples. During each iteration, one sample served as the test set, while 80% of the data from the remaining four samples constituted the training set, with the remaining 20% designated as the validation set. Each seismic event category was assigned specific numbering, as outlined in Table I.

Table I Dataset Statistics

	Training Set	Validation Set	Test Set
Blasting	1881	471	591
Collapse	1593	399	498
Nature Earthquake	2070	516	648

III. MODEL

A. Short-Time Fourier Transform (STFT)

Based on the research findings from [10] various types of seismic events exhibit distinct differences in the frequency domain, including corner frequency, bandwidth, and dominant frequency region. Therefore, the seismic signals are transformed into the frequency domain using Short-Time Fourier Transform (STFT). This approach facilitates the exploration of the frequency domain characteristics of seismic events, providing crucial information for seismic classification.

The core concept of Short-Time Fourier Transform (STFT) involves partitioning a time-domain signal into windows and performing a Fourier transform on the signal within each window. The discrete-time series representation of STFT is expressed as (1). Where $x(n)$ represents the original time-series signal, $w(n)$ represents the window function, N denotes the length of the sequence within each window, R is the gap between windows, m is the window's index, and k is the frequency index.

$$X(m, k) = \sum_{n=0}^{N-1} w(n)x(n+mR)e^{-j2\pi kn/N} \quad (1)$$

Fig. 1 depicts the processed waveforms of natural earthquakes, collapses, and explosions, alongside their corresponding short-time spectrograms. The horizontal axis represents time, the vertical axis represents frequency, and brightness signifies frequency amplitude. Distinctions in the short-time spectrogram among different seismic events are evident: natural earthquakes exhibit high amplitudes in the low and mid-frequency regions, collapses are predominantly concentrated in the low-frequency range, while explosions manifest in the high-frequency range. This illustration underscores the utility of STFT in the classification and identification of diverse seismic events.

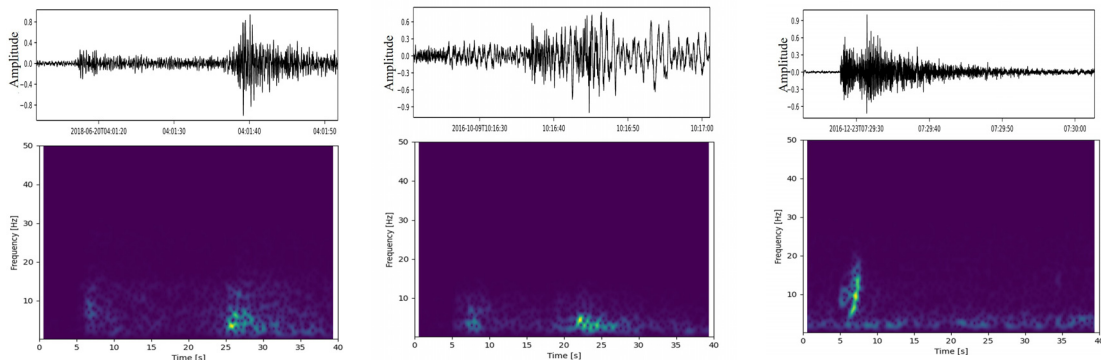


Figure 1. Waveforms and Corresponding Short-Time Spectrograms of Different Seismic Events

B. Skeletal Model

The skeletal model employed in this study is based on Xception [11]. The Xception model comprises input, middle, and output layers, totaling 14 modules. Notably, each module from M1 to M14 includes residual structures, establishing linear connections with shallow features. This effectively alleviates potential issues of gradient vanishing and exploding in the network. Additionally, each module employs depth-wise separable convolution structures, integrating considerations of channel and spatial correlations. This not only reduces the number of parameters but also alleviates computational burden.

C. Spatial Pyramid Pooling Module

Seismic waveform data typically encapsulate multiscale information, with different wavelengths of vibrations carrying distinct key information. Traditional convolutional neural networks often struggle to effectively extract these multiscale features. However, the Space Pyramid Pooling (SPP) structure, as proposed by He Kaiming and others [12], addresses this limitation. The SPP structure, achieved through operations on multiple receptive fields of different sizes, facilitates the efficient aggregation of information from various scales, comprehensively capturing the features of seismic waveforms. In addition, SPP can aggregate the feature maps of the convolutional output into a fixed-size feature vector, as illustrated in the Fig. 2, avoiding the computational and parameter complexity introduced by numerous fully connected layers.

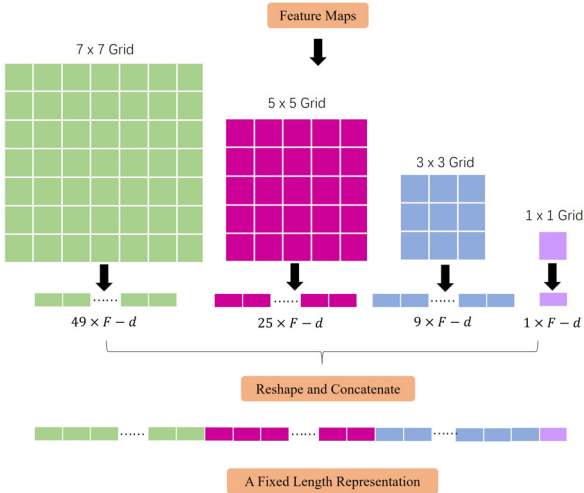


Figure 2. Diagram of Spatial Pyramid Pooling (SPP) Layer

D. Attention Module

Accurate feature extraction from seismic time-frequency images is vital for earthquake event classification. CBAM (Convolutional Block Attention Module) [13] proposes a

method to adaptively learn channel attention and spatial attention weights for each convolutional layer, based on the processing mechanism of the human visual system. This enables the network to effectively identify and focus on the most informative feature channels and spatially correlated features in seismic event classification.

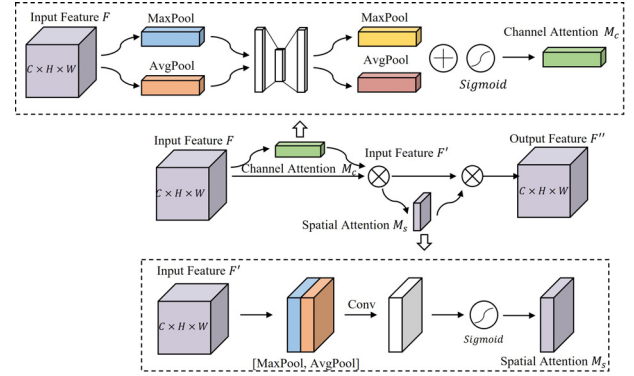


Figure 3. Diagram of Convolutional Block Attention Module

Based on the outlined architecture, Fig. 4 illustrates the network structure of our approach. The Xception model serves as the backbone network, initially converting the waveform into short-time spectrograms using the short-time Fourier transform to extract crucial frequency domain features necessary for seismic classification. Subsequently, integrating attention mechanisms prioritizes significant seismic features, thereby enhancing the model's accuracy. Lastly, spatial pyramid pooling effectively captures seismic features across diverse scales and resolutions, thereby bolstering the model's understanding and generalization capability for precise seismic event classification.

IV. EXPERIMENTAL EVALUATION

A. Experimental Setup

The hardware environment for this study is as follows: the CPU is an Intel Xeon Gold 6226R, the GPU is an Nvidia GTX 3090, and the memory size is 256GB. The software platform includes Ubuntu 18.04 as the operating system, Linux kernel version 5.4, and Python 3.8 with PyTorch 1.11 as the framework for network construction and training.

The experiment employed categorical cross-entropy as the loss function, and the convolutional neural network was trained using the Adaptive Moment Estimation (Adam) optimization algorithm with a learning rate of 0.001. The batch size was set to 32.

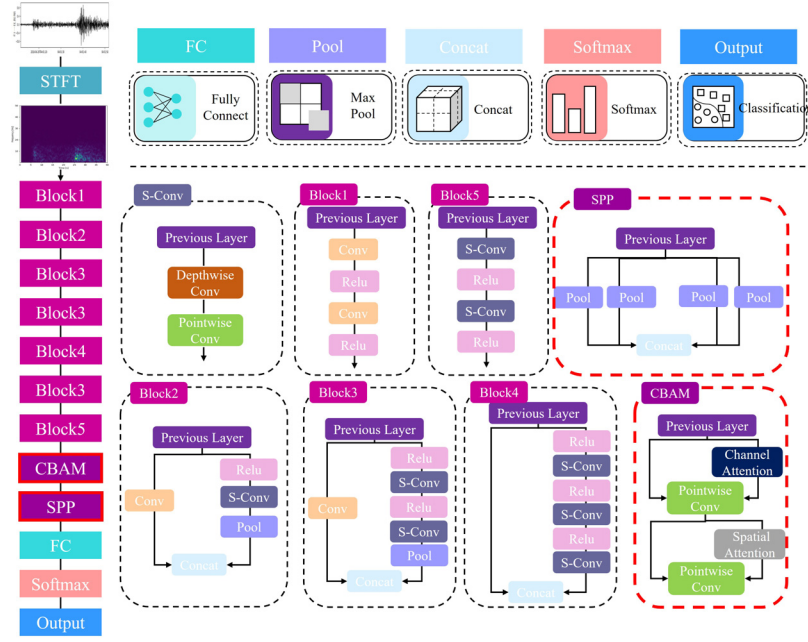


Figure 4. Proposed deep network model

B. Evaluation Indicators

To further analyze the classification performance of the proposed method, we employed six performance metrics, namely Accuracy (Acc), Sensitivity (Sn), Specificity (Sp), Precision (Pr), F1-score (F1), and Matthew's Correlation Coefficient (MCC).

$$Acc = \frac{T_p + T_n}{T_p + T_n + F_p + F_n} \quad (2)$$

$$Sn = \frac{T_p}{T_p + F_n} \quad (3)$$

$$Sp = \frac{T_n}{T_n + F_p} \quad (4)$$

$$Pr = \frac{T_p}{T_p + F_p} \quad (5)$$

$$F1 = \frac{2 \times T_p}{2 \times T_p + F_p + F_n} \quad (6)$$

$$MCC = \frac{(T_p \times T_n) - (F_p \times F_n)}{\sqrt{(T_p + F_p) \times (T_p + F_n) \times (T_n + F_p) \times (T_n + F_n)}} \quad (7)$$

In the above equations, true positives (T_p) and false positives (F_p) represent the number of correctly and incorrectly classified positive samples, respectively. True negatives (T_n) and false negatives (F_n) indicate the number of correctly and incorrectly classified negative samples, respectively.

C. Experimental Comparative Analysis

The study adopts a five-fold cross-validation strategy, training the network for 500 epochs on the training data. At the end of each epoch, the model's performance is assessed using the validation set. The model achieving the highest accuracy on the validation set is retained and subsequently evaluated on the test set. Performance metrics, including accuracy, sensitivity, and specificity, are computed after each fold, and the average values across the five-fold cross-validation are reported. The research comprises ablation experiments to analyze the contributions of individual components in the proposed model and comparative experiments to evaluate popular methods in the field, thereby assessing the proposed model's strengths and weaknesses.

1) Ablation Experiments

In this experimental study, we systematically evaluate the performance of our proposed model, denoted as Xception-SPP-CBAM, against two alternative models: Xception-SPP and the base skeleton model Xception. The input image size remains standardized at 299×299 pixels throughout the training process. Notably, our proposed method achieves substantial classification accuracies of 93.52%, 94.34%, and 94.80% for natural earthquakes, explosions, and collapses, respectively. The confusion matrix is used to compare the performance of all models, and the specific results are shown in Figure 4.

Table II Evaluation Metrics for Ablation Experiments

	<i>Acc</i>	<i>Sn</i>	<i>Sp</i>	<i>Pr</i>	<i>F1</i>	<i>MCC</i>	<i>LP/M</i>	<i>TrT/min</i>	<i>TeT/ms</i>
Xception	91.02%	91.07%	95.49%	90.94%	91.00%	86.49%	21.8	218	5.35
Xception-SPP	93.26%	93.21%	96.59%	93.28%	93.23%	89.86%	21.8	224	6.45
Our Method	94.18%	94.10%	97.05%	94.22%	94.15%	91.24%	22.2	231	6.76

LP: Learnable Parameters (M), TrT: Training Time (min), TeT: Single Image Inference Time (ms)

Table III Evaluation Metrics for Comparative Experiments

	<i>Acc</i>	<i>Sn</i>	<i>Sp</i>	<i>Pr</i>	<i>F1</i>	<i>MCC</i>	<i>LP/M</i>	<i>TrT/min</i>	<i>TeT/ms</i>
Wei et al.[7]	90.20%	90.29%	95.03%	90.33%	90.23%	85.36%	23.5	222	6.68
Zhou et al.[14]	90.72%	91.12%	95.14%	90.36%	90.71%	85.80%	9.9	169	2.77
Gao et al.[15]	91.18%	91.16%	95.54%	91.25%	91.21%	86.77%	57.01	61	1.36
Li et al.[8]	91.48%	91.65%	95.72%	91.38%	91.50%	87.21%	11.5	254	6.21
Zhang et al.[16]	92.90%	93.20%	96.32%	92.62%	92.89%	89.15%	18.5	263	6.99
Our Method	94.18%	94.10%	97.05%	94.22%	94.15%	91.24%	22.2	231	6.76

LP: Learnable Parameters (M), TrT: Training Time (min), TeT: Single Image Inference Time (ms)



Figure 5. Confusion Matrix

Table II displays evaluation metrics derived from confusion matrices, facilitating comparison. Our method achieves exceptional performance, with 94.18% accuracy, surpassing Xception-SPP (93.26%) and Xception (91.02%). Notably, our method excels in sensitivity (94.10%), specificity (97.05%), and precision (94.22%), outperforming other models. It also secures the highest F1 score (94.15%) and Matthews correlation coefficient (MCC) (91.24%), indicating superior generalization capability.

Table II also provides information on learnable parameters and runtime for each model. Despite having the most learnable

parameters, our model's introduction of the SPP structure doesn't increase this count. Moreover, the attention mechanism marginally boosts parameters by 1.83% while significantly enhancing performance. Regarding runtime, all models perform comparably. The model proposed in this article exhibits the shortest training time, totaling 218 minutes. During the evaluation process, a single sample requires 5.35 milliseconds for the skeletal model. Upon the introduction of the spatial pyramid pooling structure and attention mechanism, the evaluation time experiences a minimal increase, now standing at only 1.41 milliseconds.

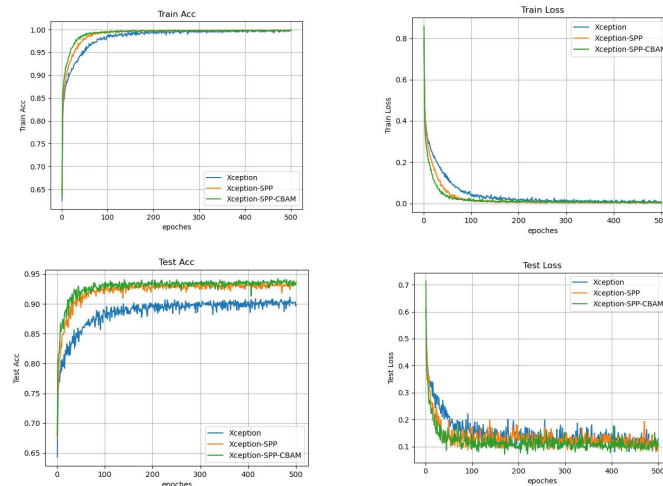


Figure 6. Accuracy Curve and Loss Curve

The accuracy and loss curves for the training and validation processes are illustrated in Fig. 6. The horizontal axis represents the training epochs, while the vertical axis indicates the metrics for accuracy and loss. It is evident that the incorporation of the SPP structure and attention mechanism significantly accelerates the overall network convergence speed, accompanied by a notable improvement in accuracy. These results underscore the substantial enhancement in both classification accuracy and convergence speed, affirming the efficacy and promise of our approach for precise classification and monitoring of seismic events.

2) Comparative Experiment

To explore the classification advantages of this model, this study conducted a comparative analysis with existing research in the field of earthquake event classification using the same dataset, as illustrated in Table III.

Table III summarizes various seismic event classification methods proposed by researchers. Wei Yonggang et al.[7] utilized a deep residual network with waveform power spectrum input, achieving 90.20% accuracy. Zhou Shaohui et al.[14] selected AlexNet for classification, reaching 90.72% accuracy. Gao Yongguo et al.[15] compared Inception V1 and SPCCNN models for binary classification, with Inception V1 achieving 91.18% accuracy. Bingjun Li et al.[8] employed STFT and LeNet5, achieving accuracies between 91.43% to 91.71%. Zhang Fan et al.[16] transformed waveform signals into log-amplitude spectrogram distribution maps, achieving 92.90% accuracy.

The experimental findings indicate that our proposed model outperforms other deep learning methods across multiple performance metrics, including accuracy, sensitivity, specificity, precision, F1 score, and the Matthews correlation coefficient. These results signify a significant advancement in seismic event classification, offering a more robust solution for seismic research and applications.

V. CONCLUSIONS

This study presents a novel approach by integrating short-time Fourier transform, attention mechanisms, and spatial pyramid pooling techniques into seismic event classification. Through five-fold cross-validation, the model proposed in this paper has exhibited enhanced generalization ability, improved network accuracy, and efficient training performance. These advantages collectively enable a more accurate differentiation of complex seismic events, thereby providing robust support for seismic-related research and applications. Future research endeavors could delve deeper into signal characteristics across diverse geological and topographical conditions, fostering a comprehensive understanding of seismic events and yielding more precise information for practical applications.

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